

Events

140

120

100

80

60

40

20

0

Data / Pred.

1.25

1

0.75

0.5

**ATLAS** Internal  
 $\sqrt{s} = 13.6\text{TeV}, 140\text{fb}^{-1}$   
2  
HH2  
Pre-Fit

|                                                                                   |             |       |
|-----------------------------------------------------------------------------------|-------------|-------|
|  | data        | 125.0 |
|  | HH          | 0.0   |
|  | $t\bar{t}$  | 7.3   |
|  | Other top   | 0.7   |
|  | Single-H    | 10.7  |
|  | VV          | 9.2   |
|  | VVV         | 0.4   |
|  | $V\gamma$   | 23.7  |
|  | W+jets      | 2.3   |
|  | Z+jets      | 145.8 |
|                                                                                   | Total       | 200.1 |
|                                                                                   | Uncertainty |       |

$\chi^2/\text{nfd} = 12.2 / 5$   $\chi^2\text{prob} = 0.03$

XGBscore